Project Synposis

1. Project Title :

CardScout : Generative AI-Powered Credit Card Optimizer and Financial Intelligence Hub

2. Introduction & Problem Statement :

In the modern financial landscape, consumers are overwhelmed by the sheer volume of credit card options, complex reward matrices, hidden penalty clauses, and dynamic fee structures. A significant lack of financial literacy leads to consumers selecting sub-optimal credit products, missing out on potential yield (cashback/travel miles), failing to meet fee-waiver pacing targets, and inadvertently damaging their CIBIL scores due to high credit utilization. Existing solutions are largely static affiliate websites that fail to provide hyper-personalized, real-time advice based on an individual's exact spending habits and existing digital wallet.

3. Proposed Solution :

CardScout AI is a comprehensive, generative AI-driven financial advisory platform. It actively analyzes a user's exact expenditure blueprint, existing banking relationships, and preferred reward types to calculate the mathematical maximum yield achievable. By chaining Google's Gemini Large Language Model with the Tavily Web Search API, CardScout fetches live web data to generate personalized strategic roadmaps, audit hidden penalties, and provide real-time transaction routing advice.

4. Key Modules and Features :

- **Guest Scout (Financial Profiling Engine):** A multi-step UI wizard that ingests user demographics, professional profiles, estimated credit scores, and granular monthly spending habits across specific categories (e.g., Forex, Quick Commerce, Utilities).
- **Intelligence Hub & Roadmap Generator:** Utilizes the LLM to cross-reference user data with live market offerings, generating a Top-5 Podium recommendation, total annual savings metrics, and a step-by-step acquisition roadmap.
- **The Battleground:** A peer-to-peer analysis tool allowing users to pit any credit card in the market against the AI's top recommendation for a comparative verdict.
- **The Obsidian Vault (Digital Wallet & Analytics Dashboard) :**

  The Obsidian Vault serves as the secure, post-acquisition command center for users to manage their credit portfolio. Backed by a PostgreSQL database, it stores authenticated user profiles, digital wallet inventories (card names and credit limits), and active subscriptions. The Vault is divided into 7 distinct analytical engines:

  1. Fee Waiver Pacing (Predictive Spend Tracker) :

- **Logic:** Actively tracks if a user is spending enough to qualify for a card's annual fee waiver. The AI fetches the exact spend target (e.g., ₹2,00,000).
- **Mathematical Modeling:** It divides the remaining required spend by the months left until renewal to generate a "required monthly run-rate".
- **CIBIL Protection:** Crucially, the system calculates the *Utilization Ratio* by comparing the required monthly spend against the user's saved credit limit. If chasing the fee waiver requires utilizing more than 30% of the card limit, the AI issues a strict "🔴 VERDICT: HOLD" warning to prevent the user from damaging their credit score.

2. Live Offers Radar (Real-Time Scraping):

- **Logic:** Replaces static affiliate websites by performing real-time web scraping.
- **Coverage:** It scans the internet for live sales, discounts, and card-specific offers across 5 distinct consumer verticals: E-Commerce (Amazon/Flipkart), Food Delivery (Swiggy/Zomato), Travel (Flights/Hotels), Transit (IRCTC), and Fuel (BPCL/HPCL/IOCL).

3. Milestone Rewards (Spend-Tier Gamification):

- **Logic:** Many users miss out on hidden, high-value bonuses (like free flight tickets or hotel vouchers) because they are unaware of specific annual spend thresholds.
- **Coverage:** This engine extracts specific spend milestones from bank terms and conditions and visually tracks them as unlockable achievements.

4. Subscription Saver (Utility & Recurring Bill Optimizer):

- **Logic:** Users input their recurring digital auto-debits (e.g., Netflix, Rent, Gym) into the PostgreSQL database, routing them to specific cards.
- **Yield Calculation:** The AI calculates the exact cashback/value-back percentage the chosen card yields for that specific Merchant Category. It then compares this dynamic credit yield against a standard UPI/Debit payment (which yields ₹0), visually showing the user their automated annual savings.

5. Forex Engine (Cross-Border Optimizer):

- **Logic:** Evaluates the hidden costs of international transactions.
- **Mathematical Modeling:** The user inputs a target foreign currency (USD, EUR, GBP, AED) and budget. The engine identifies the card's base forex markup fee (e.g., 3.5% vs. 0% for Zero-Forex cards). It then mathematically applies the mandatory 18% Indian GST on that markup, and warns the user

about Dynamic Currency Conversion (DCC) fees, revealing the true hidden penalty of using the wrong card abroad.

6. Point Valuation Matrix (Arbitrage Engine):

- **Logic:** Solves the problem of "blind redemption" by fetching the exact INR conversion value of a single reward point.
- **Coverage:** It maps the user's current point balance across three distinct redemption channels: Statement Credit (Cash), Brand Vouchers, and Airmiles/Hotel Transfers. By calculating the monetary difference between the highest and lowest value channels, it identifies "Arbitrage Opportunities," explicitly warning users if they are leaving money on the table by choosing cash over travel miles.

7. Penalty Audit (Forensic Term Extraction):

- **Logic:** Scrapes and extracts the dangerous, hidden clauses buried in the card's Terms and Conditions.
- **Coverage:** Generates a highly structured, strict financial audit detailing the Annual Percentage Rate (APR), Late Fee Slabs, Over-limit Penalties, and instantaneous interest triggers on ATM Cash Advances. An integrated AI "Red Flag Advisory" summarizes the most dangerous penalty clause into a single, punchy warning.

- **CardScout Co-Pilot :**

  A conversational AI assistant (CardScout AI) that acts as a real-time spending advisor. Users can input impromptu purchases, and the AI evaluates their digital wallet to recommend the optimal card to use while strictly enforcing a 30% utilization limit rule.

5.System Architecture & Technologies :

The system is built on a robust Python architecture. The frontend leverages Streamlit, styled with custom CSS for a modern, dark-themed frosted-glass interface. The backend utilizes PostgreSQL for secure relational data storage (users, digital wallets, and active subscriptions). System architecture includes an intelligent API rotation manager (rotate_key()) that automatically switches between pooled API keys to prevent rate-limiting and ensure high availability.

6. Security and Compliance :

User profiles are secured using advanced cryptographic techniques, specifically PBKDF2-HMAC-SHA256 password hashing with unique per-user salting, ensuring absolute protection of user authentication data within the PostgreSQL database.

7. Conclusion :

CardScout AI bridges the gap between complex banking products and consumer financial literacy. By combining live data scraping, predictive mathematical modeling, and generative AI, the platform empowers users to optimize their digital wallets, maximize their returns, and avoid predatory financial penalties.

Softwares used :

- **Programming Language:** Python 3.11
- **Frontend Framework:** Streamlit (with Custom CSS)
- **Backend & Database:** PostgreSQL (hosted on Supabase), integrated via psycopg2
- **Artificial Intelligence:** Google Generative AI API (Gemini Model) with Custom Prompt Engineering
- **Real-time Data Fetching:** Tavily Web Search API (for live scraping of deals and dynamic bank terms)
- **Security & Encryption:** hashlib (PBKDF2-HMAC-SHA256 password hashing)
- **Report Generation:** fpdf library for dynamic PDF export